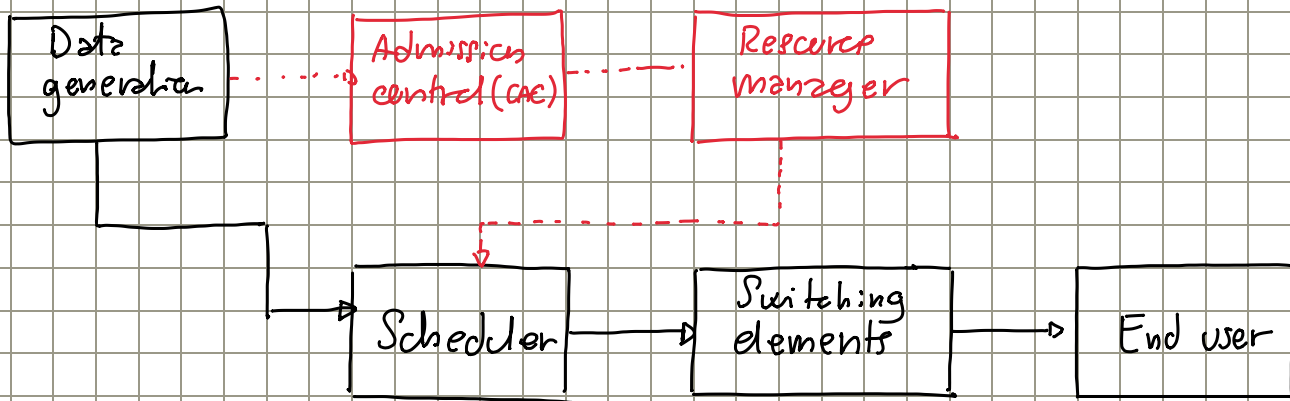


HOW TO BUILD A NETWORK MODEL

Main modules

- Data generation: traffic/source models
- Admission control (CAC): accepts or reject new flow
- Resource manager: allocate resource
- Scheduler: assigns resources to admitted flows
- Switching elements: buffers along the path
- End user: perceived QoE



We'll start with CONCEPT OF MODELING, then DATA GENERATION, QoE, RM, CAC, Scheduler and Queuing

NETWORK ABSTRACTION

A Network can be described as:

- 1) **Users**: entities that require system's services to achieve their goals
- 2) **Resources**: elements needed to provide services to users
- 3) **RM**: (missing)
- 4) **Utility** metrics: how the comb. of 1, 2, 3 creates a service for user

Users

Objective: maximize QoE $\rightarrow$ depends on target/user needs

Resources and Requests

- Resources: elements available in finite quantities used by system to provide services to users

c_k = max # of resource of type k available in the system

- Requests: number of resources of type k demanded by user s

$d_s(k)$ = # of resources of type k demanded by user s

Questions: for user, does it make any sense to have:

$$s \in D \quad d_s(0) = 10 \text{ Mbps}$$

$$d_s(1) = 6 \text{ Mbps}$$

Write explanation.

Bottleneck

$$s \in W \quad d_s(0) = 3 \text{ Mbps}$$

$$d_s(1) = 2 \text{ Mbps}$$

RM (Resource Management)

Assigns resources to user $\rightarrow$ centralised
 $\rightarrow$ distributed

$r_s(k)$ = # res of type k assigned to user s

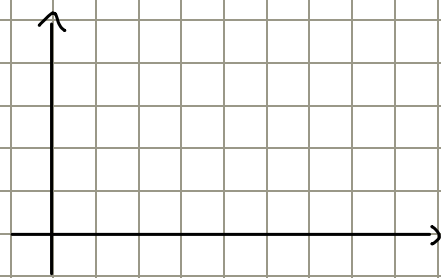
$r_s(k) = [r_s(0), \dots, r_s(M-1)]$ = resource allocation vector for user s

RM must require:

- Feasibility: $\sum_{s \in S} r_s(k) \leq c_k, \forall k \Rightarrow$ don't assign more than available

- Principle of Least Privilege (PLP): $r_s(k) \leq d_s(k) \quad \forall k \Rightarrow$ don't assign more than required

Ex



$$\hat{d}_s \in \{0, 1\}$$

$$d_s \in \{0, 1\}$$

$$IP[d_s = x \mid \hat{d}_s^1 = 1] = \begin{cases} 0.98 & x = 1 \\ 0.02 & x = 0 \end{cases}$$

$$IP[d_s = x \mid \hat{d}_s^1 = 0] = \begin{cases} 0.001 & x = 1 \\ 0.999 & x = 0 \end{cases}$$

if $C = 300$ seats

you accept exactly c reservations, what will be the average efficiency of your service?

$$\eta = \frac{\sum_{s=1}^n d_s}{C} \rightarrow E[\eta] = ? = \frac{C \cdot IP[d_s = 1 \mid \hat{d}_s^1 = 1]}{C} = 98\%$$

$\uparrow$
 $N \cdot c$

PLP most of the time is violated

A trivial RM criterion is to equally split the available resources among all users

• D: downloaders

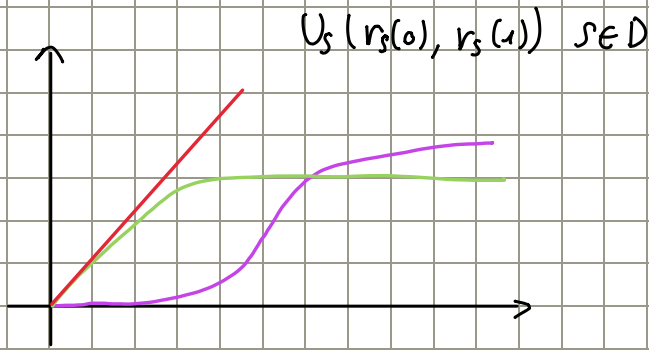
• W: watchers

$$\forall s \in D \cup W, \quad r_s(0) = \frac{C_0}{|D \cup W|}, \quad r_s(1) = \frac{C_1}{|D \cup W|}$$

Rest of examples, we need utility functions

Utility functions

- How much the resource allocation vector r_s contributes to achieve the goal of user s
- Utility can reflect different performance metrics
- Costs can be modeled as negative utility functions



System utility

Overall system cost/utility can depend on:

- Total amount of resources allocation
- Utilities experienced by customers
- Other factors

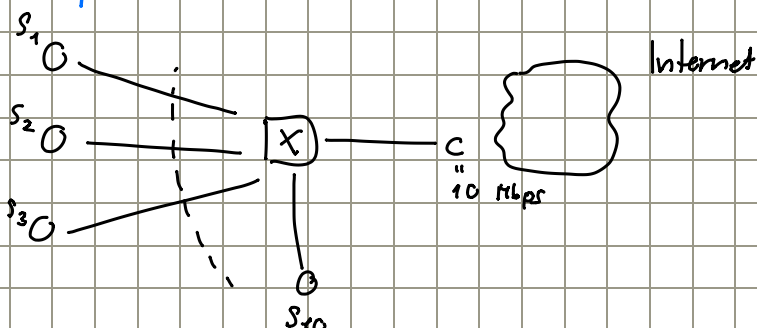
The problem

Maximize a certain objective function $\Rightarrow \max_r U_0$

Subject to constraints:

- Feasibility
- Min performance constraints

Examples



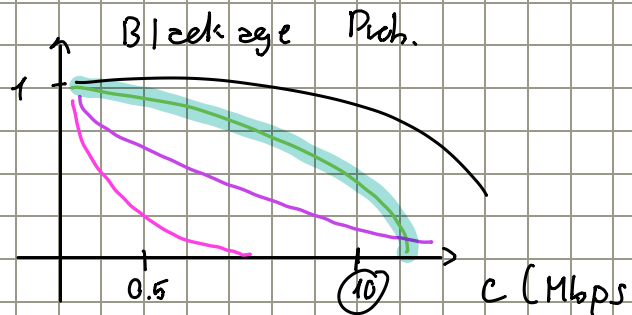
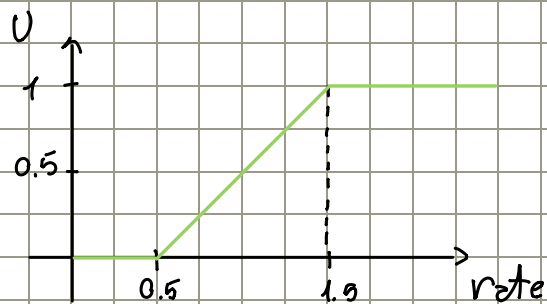
Customers = users that makes videocall

Resource types = Capacity (bitrate) shared among the users

Resource demand = depends on how much makes a videocall demands

Utility Functions = ?

Suppose that a videocal requires 1.5 Mbps while ongoing, full quality, and 0.5 Mbps at minimum quality. $\rightarrow$ this is already a utility function for my user.



NETWORK

Point process

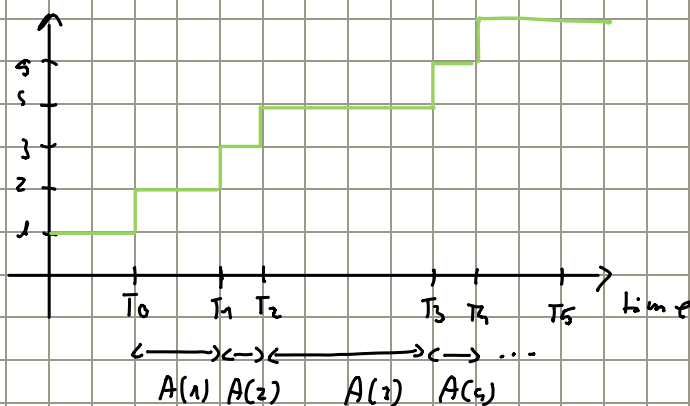
- Point process $\{T_i\}$: generation instant of the i -th "event"
 $\rightarrow$ event may consist in the generation of a single IU or a batch of IUs

- Counting process: $N(t)$ = # arrivals up to time t

$$N(t) = \max\{n: t > T_n\} = \sum_{k=1}^{\infty} 1(t > T_k)$$

- Interval process $\{A_i\}$: sequence of time intervals between consecutive arrivals

$$A_i = T_i - T_{i-1}$$



- Compound traffic $\{B_i\}$: number of IUs created at the i -th event

- Workload $\{W_i(w)\}$: amount of work brought to a system by the

i -th arriving IO at epoch n

- Aggregate offered workload $\{S_g(t)\}$: sum of the workload up to time t
- Aggregate average offered traffic: $\frac{S_g(t)}{t}$

Renewal Process

- Generalization of a Poisson Process
 - Poisson Process: $\{A_i\}$ are iid $\sim e(\lambda) \Rightarrow$ point process is a homogeneous Poisson process
 - $\{A_i\}$ are iid $\sim F(\dots) \rightarrow$ Renewal Process
 $\hookrightarrow$ CDF of $\{A_i\}$

$\{A_i\} = \text{cost} = 5$ Is it Poisson? No, because it's not exponential.
It's a renewal process!

$\{A_{i+1}\} = \begin{cases} A_i & \text{w.p. } 0.1 \\ \text{random}(0,1) & \text{w.p. } 0.9 \end{cases}$ Is it Poisson? No, same reason
Is it Renewal? No, because $\{A_i\}$'s are not iid

Renewal process: let $\{A_i\}$... write definition...

Renewal-reward Process

Let $\{G(i)\}$ be a series of iid finite r.v.s. Then process

$$Y(t) = \sum_{n=1}^{N(t)} G(n)$$

is a renewal-reward process

- $Y(t)$ is defined by $\{A_i\}$ and $\{G_i\}$
- Pairs $\langle A(i), G(i) \rangle$ must be iid, but $A(i)$ and $G(i)$ can be correlated
- In particular, $G(i)$ can be a function of $A(i)$

Elementary renewal theorem